

Prototyping, Designing with FPGA Boards Made Affordable and Absorbing



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Field programmable gate array (FPGA) development boards, like any device following Moore's law, are more powerful and inexpensive than ever before. The amount of logic you can fit on these devices is staggering, informs Mark A. Bowers, co-founder, MicroNova Electronics. He says, "These have evolved from containing simple glue logic to emulating entire processors and churning through complex digital signal processing (DSP)."

Today, developers are interested in choosing FPGAs for designing their prototypes of pre-silicon deliverables due to their moderate portability in nature, informs Sudharsan Palanikumar, senior analyst, Femto Logic Design Pvt Ltd. He says, "We also see that FPGAs are chosen for this prototyping objective for their ease-of-integration method in software

development processes for demonstration."

FPGA platform availability has improved very much in India today. There are many affordable options available, informs Anoop Jose, manager - research and development, Numato Systems Pvt Ltd. He says, "In 2008, one had no option but to import low-cost FPGA boards. And importing even the cheapest board would have costed ₹ 10,000 or more, once shipping charges and import duties were considered."

Industries use off-the-shelf FPGA development boards as proof-of-concept/prototype hardware platform to validate logic, architecture, data flow or performance requirements and eventually custom-build their own product based on FPGA/application-specific integrated circuit (ASIC). Dhiraj Kumar, director, Systems Engineering, Argus Technologies, says, "This specific

Point of view: Tips for beginners looking to buy FPGA development boards

FPGA size not that important. If you are just beginning, you will not likely be creating huge designs. Even the smallest FPGA you can find on a development board will suffice for quite some time.

SoC FPGAs not recommended. While system on chip (SoC) FPGAs are really cool, these are also fairly complicated to work with and definitely not a recommended starting point for beginners. Beginners can experiment with Mojo board, where you get to program Arduino (ATmega32U4) and the FPGA (XC6SLX9) as it aims to simplify getting started with FPGA and digital designing.

Availability of tools and support. The most important factor when looking for a development board is the tools and support offered. If either is lacking, it will make learning how to use the board so much harder. Almost all FPGA development boards are targeted at FPGA developers, not people wanting to be FPGA developers. This is incredibly important and what I feel is the biggest factor setting Mojo board apart.

Tool cost is another major factor. You do not want to drop a thousand dollars on some super-powerful board only to find out that you have to spend another chunk of money to get access to the tools to even work with it. Many larger FPGAs require a licence to work with. I highly recommend avoiding these for as long as possible. Also, some boards may require a JTAG programmer. Xilinx's programmer costs a staggering US\$ 225. Mojo has no hidden costs. All tools are free and the only thing you need besides the board is a micro universal serial bus (USB) cable.

Avoid feature overload. Many development boards are designed to show-off some features of the FPGA rather than be super useful. Too many boards are packed with useless features that take up general I/O pins and drive up the cost. The first (and only) FPGA development board I bought was Spartan-3AN evaluation board. It cost US\$ 230 and had all the features you could think of. However, it only had a handful of I/O pins that were easy to hook up to something I made. I ended up never using this board beyond playing with a few of its demonstrations.

Mojo was designed to be minimal. You have access to basically all I/O pins. If you want some extra feature, you can add it on as a shield. There is no need to jam everything on a single board. Mojo was designed to be used in bigger projects and not just to demonstrate features on the board.

FPGAs are not that complicated. FPGAs are amazing. I fell in love with these when I first learnt what these were. The idea that I could easily create my own digital circuits was captivating. However, I had a ton of help learning about these, and before I had that, I failed to do it on my own. I want to change this so that anyone who wants to learn about these can, without needing a personal mentor or going to school for electrical engineering. I think that FPGAs are seen as more complicated than these actually are, but it is just about getting the right combination of tools and tutorials to really open the doors for many people.

— Contributed by Justin Rajewski, CEO, Embedded Micro